$$= \frac{1}{5}[\log 14 - \log 9] + \frac{7}{5} \times \frac{\sqrt{5}}{3} \left[\tan^{-1} \frac{x}{\left[\frac{3}{\sqrt{5}}\right]} \right]_0^1 = \frac{1}{5} \log \frac{14}{9} + \frac{7}{3\sqrt{5}} \tan^{-1} \frac{\sqrt{5}}{3}.$$

Example 9.7

Evaluate : $\int_0^1 [2x]\,dx$ where $[\cdot]$ is the greatest integer function.

Solution

$$\int_0^1 [2x]\,dx = \int_0^{\frac{1}{2}} [2x]\,dx + \int_{\frac{1}{2}}^1 [2x]\,dx = \int_0^{\frac{1}{2}} 0\,dx + \int_{\frac{1}{2}}^1 1\,dx = 0 + [x]_{\frac{1}{2}}^1 = 1 - \frac{1}{2} = \frac{1}{2}.$$

Example 9.8

Evaluate : $\int_0^{\frac{\pi}{3}} \frac{\sec x \tan x}{1 + \sec^2 x}\,dx$.

Solution

Let $I = \int_0^{\frac{\pi}{3}} \frac{\sec x \tan x}{1 + \sec^2 x}\,dx$. Put $\sec x = u$. Then, $\sec x \tan x\, dx = du$.

When $x = 0$, $u = \sec 0 = 1$. When $x = \frac{\pi}{3}$, $u = \sec \frac{\pi}{3} = 2$.

$$\therefore I = \int_1^2 \frac{du}{1+u^2} = [\tan^{-1} u]_1^2 = \tan^{-1}(2) - \tan^{-1} 1 = \tan^{-1}(2) - \frac{\pi}{4}.$$

Example 9.9

Evaluate : $\int_0^9 \frac{1}{x + \sqrt{x}}\,dx$.

Solution

Let $\sqrt{x} = u$. Then $x = u^2$, and so $dx = 2u\,du$.

When $x = 0$, $u = 0$. When $x = 9$, $u = 3$.

$$\therefore \int_0^9 \frac{1}{x+\sqrt{x}}\,dx = \int_0^3 \frac{1}{u^2+u}(2u)\,du = 2\int_0^3 \frac{1}{1+u}\,du = 2\left[\log|1+u|\right]_0^3 = 2[\log 4 - 0] = \log 16.$$

Example 9.10

Evaluate: $\int_1^2 \frac{x}{(x+1)(x+2)}\,dx$.

Solution

Let $I = \int_1^2 \frac{x}{(x+1)(x+2)}\,dx$.

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